Minimax Optimal (Parks-McClellan) FIR Filter Design

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) The minimax criterion finds the filter that minimizes the worst case error between the desired and actual filter response.

2) A weighting function is used to place more importance on the error in some frequency bands than others.

3) Parks and McClellan developed an algorithm for solving the minimax filter design problem.

    a) The algorithm is iterative.

b) It uses the alternation theorem which characterizes the error of polynomial approximations.

c) Minimax designs are equal ripple. The error in the pass and stop bands alternates in sign within a constant range.

4) if the weight in the passband is one, then the weight in the stop band is set to the ratio of the desired ripple in the passband to the desired ripple in the stop band.

5) Guess-and-check is normally used to find the filter order that satisfies the design constraints.

Video Screenshots

0:20

Type I FIR: M even, even symmetry $h[n] = h[M-n]$ /

$$H(e^{j\omega}) = e^{-j\omega M/2} \underbrace{\sum_{k=0}^{M/2} a[k]\cos(\omega k)}_{A(\omega)}$$

$$a[0] = h[M/2]$$
$$a[k] = 2h[\tfrac{M}{2}-k], \; k=1,\ldots,\tfrac{M}{2}$$

Minimax error design

$$\min_{a[k]} \max_{\omega \in F} \underbrace{\{W(\omega)|H_d(\omega) - A(\omega)|\}}_{\varepsilon(\omega)}$$

$H_d(\omega)$: real-valued desired response

$|H_d(\omega) - A(\omega)|$: approximation error at ω

$W(\omega)$: non negative weighting function

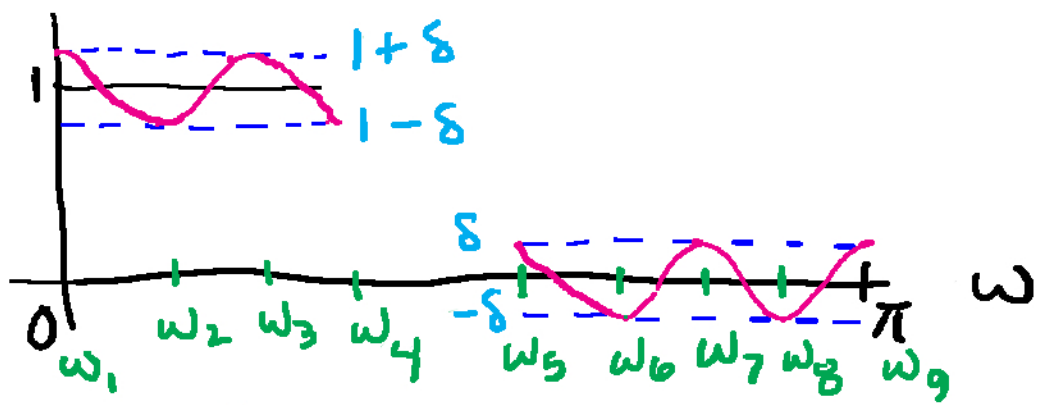
Alternation Theorem ($L = M/2$)

At least $L+2$ extremal frequencies $\omega_k, k=1,2,\cdots,L+2$

1) error alternates between two equal maxima and minima: $\varepsilon(\omega_k) = -\varepsilon(\omega_{k+1})$, $k=1,2,\cdots L+1$

2) error at ω_k equals maximum absolute error

$$\delta = |\varepsilon(\omega_k)| = \max_{\omega \in F} |\varepsilon(\omega)|$$



Design approach: find extremal frequencies ω_k, choose $a[k]$ to minimize error at ω_k

Parks McClellan Algorithm

1) Guess at initial extremal frequencies $\omega_1, \omega_2, \ldots \omega_{L+2}$

2) Find $a[k]$, δ to satisfy alternation criterion

$$(-1)^k \delta = W(\omega_k)\left[H_d(\omega_k) - A(\omega_k)\right] \Rightarrow A(\omega_k) + \frac{(-1)^k \delta}{W(\omega_k)} = H_d(\omega_k)$$

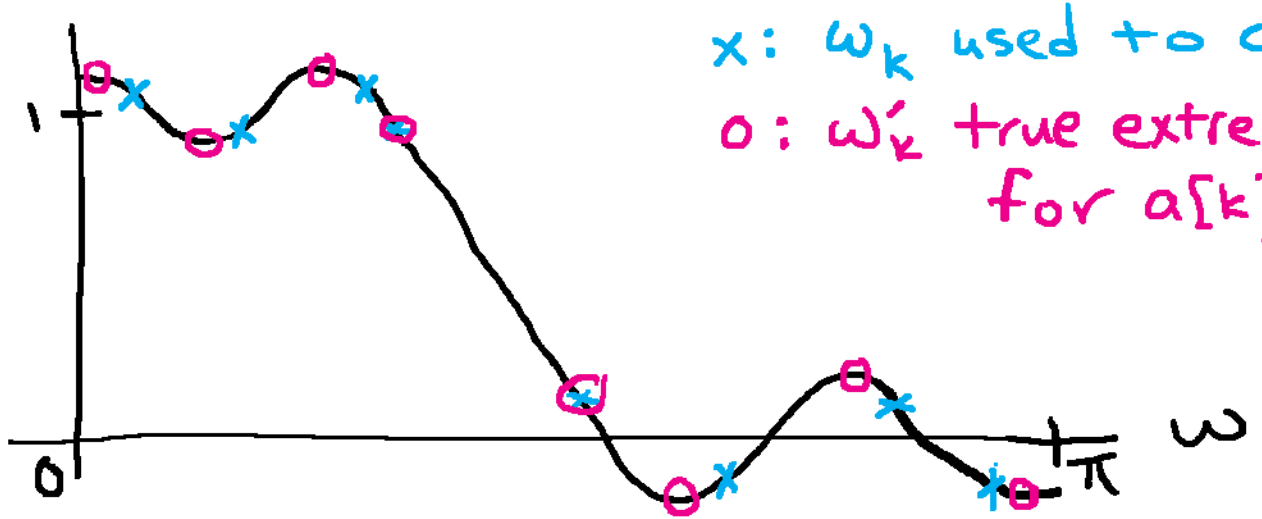
$$\begin{bmatrix} 1 & \cos\omega_1 & \cos 2\omega_1 & \cdots & \cos(L\omega_1) & \frac{+1}{W(\omega_1)} \\ 1 & \cos\omega_2 & \cos 2\omega_2 & \cdots & \cos(L\omega_2) & \frac{-1}{W(\omega_2)} \\ & \vdots & & & & \\ 1 & \cos\omega_{L+2} & & \cdots & \cos(L\omega_{L+2}) & \frac{(-1)^{L+2}}{W(\omega_{L+2})} \end{bmatrix} \begin{bmatrix} a[0] \\ a[1] \\ \vdots \\ a[L] \\ \delta \end{bmatrix} = \begin{bmatrix} H_d(\omega_1) \\ H_d(\omega_2) \\ \vdots \\ H_d(\omega_{L+2}) \end{bmatrix}$$

Parks McClellan Algorithm

1) Guess at initial extremal frequencies $\omega_1, \omega_2, \ldots \omega_{L+2}$

2) Find $a[k]$, δ to satisfy alternation criterion

3) Compute true alternation frequencies ω'_k for $a[k]$



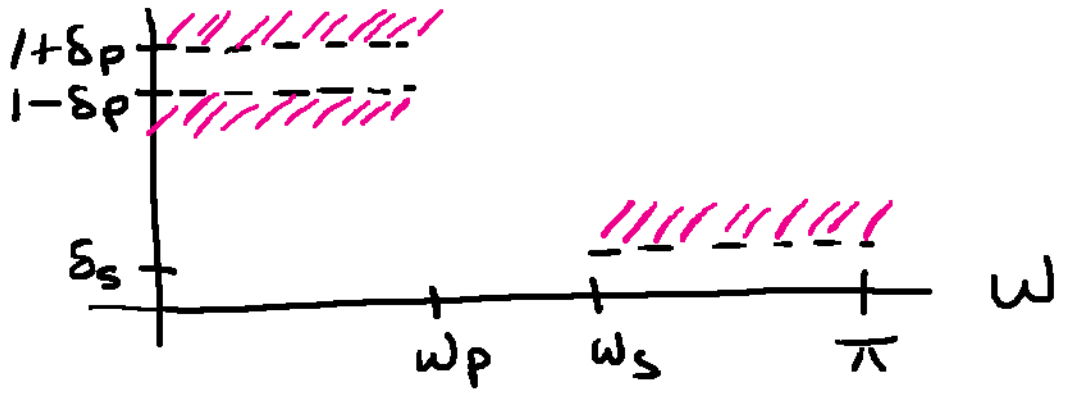
x: ω_k used to compute $a[k]$

o: ω'_k true extremal frequencies for $a[k]$

4) If $\omega'_k \neq \omega_k$, set $\omega_k = \omega'_k$ and go back to step 2.

Choosing W(ω)

$$W(\omega) = \begin{cases} 1 & 0 \le \omega \le \omega_p \\ \delta_p/\delta_s & \omega_s \le \omega \le \pi \end{cases}$$



Larger weight on bands with tighter tolerance

Choosing M

(lowpass) $$M \approx \frac{-10 \log_{10}(\delta_p \delta_s) - 13}{2.324 (\omega_s - \omega_p)}$$

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction